Objectives

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1. Students will be able to characterize the autonomic processes of the GU system and how they affect the physiology of a patient. They will also be able to discuss the location of related viscerosomatic reflexes as they relate to the GU system.
2. Students will be able to demonstrate knowledge of the lymphatic, fascial, visceral and musculoskeletal structures related to the GU system; and how changes to these structures may affect the function of GU system.
3. Students will the be able to identify and describe the location of Chapman's points related to the GU system and recognize how these points are used to diagnose and treat GU problems.
4. Students will be able to determine the goals for treating patients with osteopathic principles and practices.
5. Students will be able to apply the 5 models of Osteopathic care and treatment for GU disorders.

Neurological

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- Facilitated spinal segments may be a result of viscerosomatic reflexes

- Efferent system
  - “Autonomic Nervous System” (ANS) sympathetic and parasympathetic nerves innervate the viscera and somatic systems

- Afferent system
  - Visceral nerves carry impulses from the organ to the spinal cord (SC)
  - Usually the nerves use the same pathways as an organ’s sympathetic innervation
  - These nerves synapse in SC at the organ level of sympathetic innervation

- Facilitation of a spinal segment creates increased activity at the synapses in the SC level
  - The threshold level is lowered and easier to stimulate, resulting in discharge of impulses.
  - These impulses and synaptic mediators (NE, ACH) alter the configuration of the spinal vertebrae, hence somatic dysfunction.

- Facilitated SC segments may fire autonomic outbursts to related organ and somatic structures.

Neurological

Viscerosomatic and Somatovisceral

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- Visceral dysfunction may result in somatic dysfunction, and vice versa.
- Improvement of visceral function may improve somatic function.
- Treatment of somatic dysfunction may improve the related visceral function.
- “Cross talk” between the visceral and somatic systems at the spinal cord level provides diagnostic clues of visceral dysfunction as they present as somatic dysfunctions.
- Treat the musculoskeletal system and facilitated spinal segments to affect organ autonomic hyperactivity

Neurological

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Sympathetic

T10 - T11:
- Kidneys
- Proximal ureters
- Testes and ovaries

T12 – L1:
- Distal (lower) ureters

T12-L2:
- Bladder
- Uterus

T12 – L2:
- Prostate
- Genital cavernous tissue

Parasympathetic

Vagus nerve:
- Kidneys
- Proximal (upper) ureters
- Testes/Ovaries

Remember to treat the occipital-mastoid suture, C1 and C2 for vagus innervation

Sacral nerves S2, S3, S4 (Pelvic Splanchnic Nerves):
- Distal ureters
- Bladder
- Prostate
- Uterus
- Genitalia

Neurological

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- Chapman's reflex points are "Gangliform Contractions" or "Reflex points" that resolve with treatment
  - "Thickened and sensitive to heavy pressure"
  - They are about the size of a BB or a pearl of tapioca or an overcooked butterbean, severe visceral involvement

- Treat the Pelvis First : "your first act should be to replace the innominate"
- Then treat all significant anterior Chapman's points
- Lastly treat all posterior Chapman's points

- Treatment method is a rotary motion with one finger, usually for approximately 10 to 15 seconds

- "Do not use excessive pressure on either anterior or posterior points. You will get results more quickly and less painfully by gentle rather than forceful means."Frank Chapman, D.O.

Biomechanical

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- Fascia has proprioceptive and nociceptive properties and is able to contract and relax according to its input.
- Fascial contraction may lead to spasm and pain, which can be relieved with OMT.
- Due to the body's interconnected fascial connections, pelvic diaphragm dysfunction may affect the respiratory diaphragm function
- The posterior diaphragmatic fascia has retroperitoneal connections.
- Directly relevant for the GU system include the psoas muscle, quadratus lumborum and the kidneys which are joined together by an "interfascial plane."
- Techniques range from muscle energy, counterstrain, soft tissue and myofascial release of the posterior back musculature.

Biomechanical: Renal Fascia

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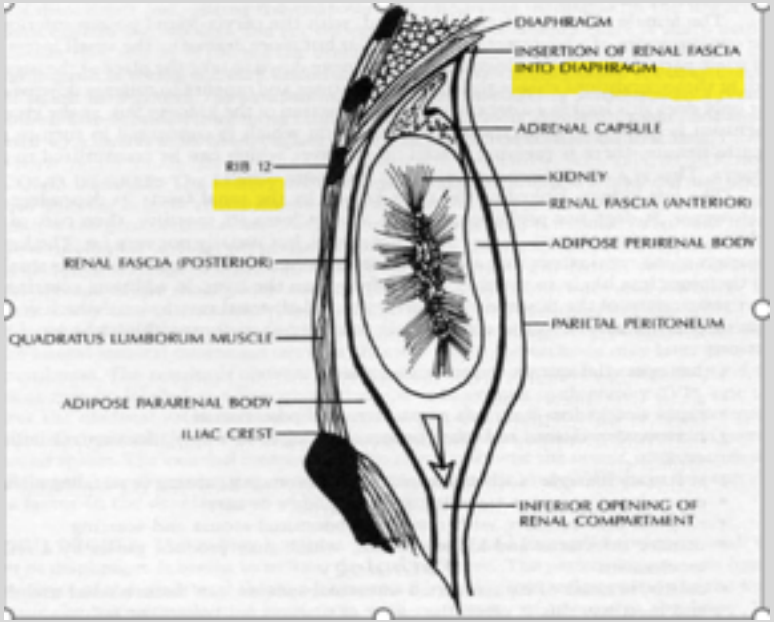
Renal Fascia Inserts into the Diaphragm

Treating the iliopsoas, quadratus lumborum and the 12th rib can directly affect the fascia around the kidney.

Both the psoas and quadratus lumborum border the perirenal fascia.

Gerota's fascia is another name for perirenal fascia surrounding the kidney.

The renal fascia allows for renal mobility of about 3 cm and varies with breathing. Renal motion can be affected by cardiac conditions such as congestive heart failure as the heart has fascial connections to the diaphragm.



Biomechanical: Ureter

The ureter is a retroperitoneal muscular organ joining the kidney to the bladder.

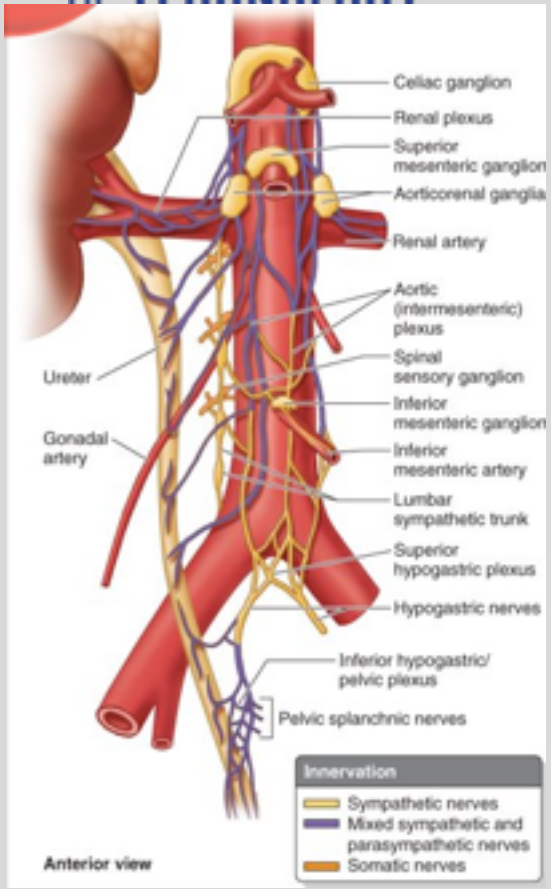
The peri-ureteric fascia is a continuation the renal fascia.

The ureter courses along the iliopsoas muscle and fascia and anterior to the bifurcation of the common iliac artery.

There are certain locations for a high chance for renal calculi obstruction at the ureteropelvic and ureterovesicular junctions

Due to fascial connection TREAT the Iliopsoas for renal stones

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Behavioral

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- Psychological states of anxiety, stress and depression
- Psychosocial environment: healthy work, family, and daily living, etc.
- Healthy lifestyle: proper diet, exercise and sleep behaviors

- Consider how osteopathic care and OMT affects the immune system with simple techniques to treat lymphatic flow and how this can alter psychological processes, psychosocial interaction and life style modifications.

Behavioral

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Consider how addressing your patients with the osteopathic behavioral model can help to ameliorate patient's concerns and sensitivities with respect to conditions such as urinary incontinence, fertility, pregnancy, personal and gender identity, sexual and physical trauma, mood, erectile dysfunction, pelvic inflammatory disease, endometriosis, etc.

Metabolic

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- Remember that when addressing a patient's medical needs, it is not only about getting the right test, using the correct medications, but also counseling and educating the patient on proper nutrition and exercise.

- FOOD: 5-9 servings of fresh fruits and vegetables per day

- One cup of leafy green vegetables is one serving.

- A half a cup of chopped vegetables and fruit is one serving.

- Remind the patient to stay well hydrated with water instead of sugary drinks and juices.

Instead of focusing on avoidance consider what the patient can and is willing to do to stay healthy!

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Goals of Treatment

- Treat facilitated segments to normalize sympathetics (T10-L2)
- Normalize parasympathetics by treating the sacrum and vagus
- Improve the lymphatic flow by treating diaphragms
    - OA, thoracic inlet, respiratory diaphragm, pelvic diaphragm, inguinal region
- Treat thoracic, lumbar, sacrum, rib cage, respiratory diaphragm, innominates/ pubic bones, and pelvic diaphragm
- Chapman's Points: treat pelvis, then the anterior points, next posterior points

Case 1: HPI

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A 26-year-old male presents to your office complaining of (c/o) fevers, chills, muscle aches, dysuria, frequent urination, cloudy urine and discomfort when sitting. He says that when he urinates he feels like he can’t get it all out. The patient denies blood in the urine. The patient had a cystoscopy (scope that passes through the urinary tract into the bladder) three days ago due to enlarged prostate and urine retention. No prostate biopsy was indicated.

Case 1: HPI

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Allergies: NKDA

Medications: None

Past Medical History: Denies

Past Surgical History: Denies

Family History: Both parents alive and well. No other significant family history

Social History: Denies alcohol and tobacco use. Uses cocaine on weekends. Is sexually active with different multiple partners on weekends when using cocaine.

Case 1: ROS

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Constitutional: Fevers, chills, generalized muscle pains

HEENT: Occasional tension headaches, no headache today

GI: Generalized lower abdominal and pelvic pain

GU: As per HPI. Painful ejaculation. No blood or puss in the urine.

Musculoskeletal: Pain in the buttocks as above

All other systems negative

Case 1: Physical Exam

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Vitals: Temp: 100.1, Pulse: 98, BP: 124/78, R: 12,
PulseOx: 98% , Ht: 5ft 11in; Wt: 220 BMI 30.7

General: NAD, A&O x 4, Obese
HEENT:Normocephalic,atraumatic (NCAT)

Abdomen: + Bowel sounds, no guarding/rigidity/rebound, + suprapubic tenderness
Rectal exam: good sphincter tone, positive for prostate tenderness and bogginess, negative stool guaiac, no rectal masses
Musculoskeletal: Normal thoracic kyphosis and cervical lordosis. +Mild costovertebral tenderness to percussion. Normal gait. Muscle strength 5/5 in all 4 extremities
Neuro: CN II-XII in tact, no other gross neurological deficits

Osteopathic Structural Exam:
Thoracic and Lumbar: T11 – L2 with red reflex.
Sacrum: S2-4 boggy and tender. Bilateral sacral flexion. Medial S2 tenderpoint
Pelvis: Right inflare innominate
Visceral: Chapman's reflex along the lateral thighs worse on the right.

Case 1: Questions

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- What is on your differential diagnosis?
- What tests could you immediately order?
- Besides treating medically, what are your treatment goals in terms of the five models of Osteopathic care?
- What OMT might you perform?

Case 1: Questions

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- Are there any contraindications to Osteopathic Manipulative Treatment and are there any treatments that you may want refrain from initially?
- Are there any Osteopathic Treatments that you could implement right away?
- How would you address the sympathetics?
- How would you address the parasympathetics?
- How would you address the lymphatics?
- What other regions would you address?

Case 2: HPI

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A 48 year old female presents to the office with fever, chills, muscle aches, pain, frequent urination, and right low back pain. She says that it burns when she urinates and she feels like she can't get it all out.

O: 3 weeks ago began as burning on urination, now worsening. Back pain, abdominal pain, fever, chills began 3 days ago
P: Nothing makes it better or worse
Q: dysuria is burning; back pain is sharp
R: no radiation
S: dysuria is 5/10; back pain is 8/10
T: constant pain with urinating; back pain is worse with movement

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Case 2:

Allergies: Tylenol
Medications: Hydrochlorothiazide
PMH: HTN
PSH: Denies
FH: Denies
SH: denies alcohol and drug use. Smokes 1 pack cigarettes per day. Sexually active with one partner.

ROS:
Constitutional: Fever, chills, muscle aches
GI: generalized lower abdominal pain, occasional nausea with pain in the back
GU: as per HPI
All other systems negative

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Case 2

How would you address sympathetics?

- Kidneys, proximal ureters: T10-T11
- Distal ureters: T12-L1
- Bladder: T12-L2

How would you address parasympathetics?

- Kidneys, proximal ureters: Vagus nerve
- Distal ureters, Bladder: Pelvic Splanchnic Nerves (S2, S3, S4)

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Case 2

How would you address lymphatics?

- Kidneys and upper ureter drain into the Lateral Aortic Nodes
- Lower Ureter drains into the common, internal, or external iliac nodes
- Bladder drains into the external iliac nodes
- Urethra drains into internal iliac nodes